

TITLE PAGE

- **Problem Statement ID** – Question 4
- **Problem Statement Title-** AI-Powered Behavioral Anomaly Detection for Cybersecurity
- **Theme-** Cybersecurity - Industrial-Edge Security
- **PS Category-** Software
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- **Student ID :** 1BM24AI401

IDEA TITLE

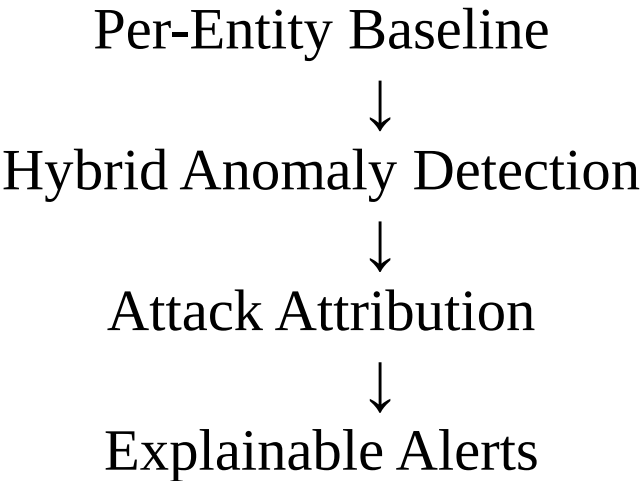
Objective:

- Learn per-entity normal behaviour
- Detect, classify & explain cyber threats
- Real-time, streaming anomaly detection

Challenges Addressed:

- Novel & Zero-Day Attacks
- Low-and-Slow Threats
- Low False Positives
- Imbalanced Security Data (~1% attacks)

Core Approach:



Key Innovations:

- Isolation Forest + GRU (Hybrid Detector)
- Novel Attack Class
- SHAP + Counterfactual Explanations
- Streaming-Ready Incremental Features
- Risk Calibration & Alert De-duplication

TECHNICAL APPROACH

Technologies

- Python, NumPy, Pandas
- Scikit-learn (Isolation Forest, Random Forest)
- PyTorch (GRU Autoencoder)
- SHAP
- Streamlit

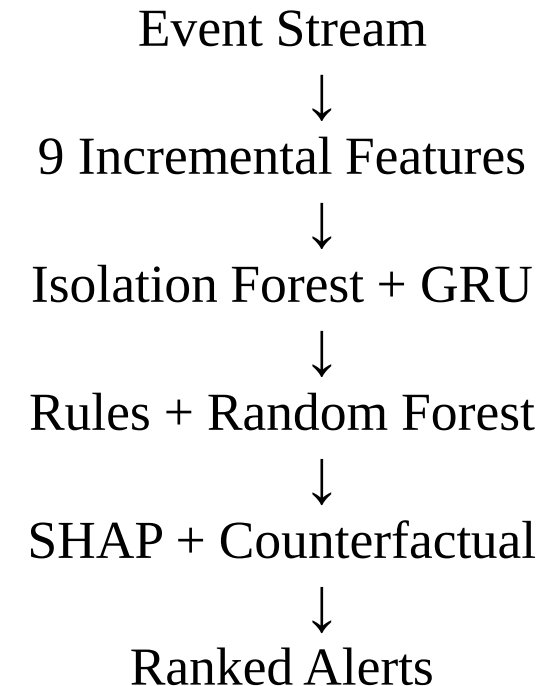
Performance

- PR-AUC: 0.42
- Recall@2% Budget: 67%
- Attack Naming: 100%
- Brier Score: 0.007

Model Characteristics:

- Unsupervised & Label-Blind Detection
- Per-Entity Feature Engineering
- Streaming-Ready Architecture
- Temporal Train-Test Split

Pipeline:



FEASIBILITY AND VIABILITY

Feasibility:

- ~75 s end-to-end pipeline execution
- GRU inference in sub-ms/event
- Incremental feature engineering
- Streaming-ready (Redis state caching)

System Robustness:

- Cold-Start aware detection
- Alert De-duplication
- Calibrated Risk Ranking
- Real-Time Streaming Support

Practical Impact:

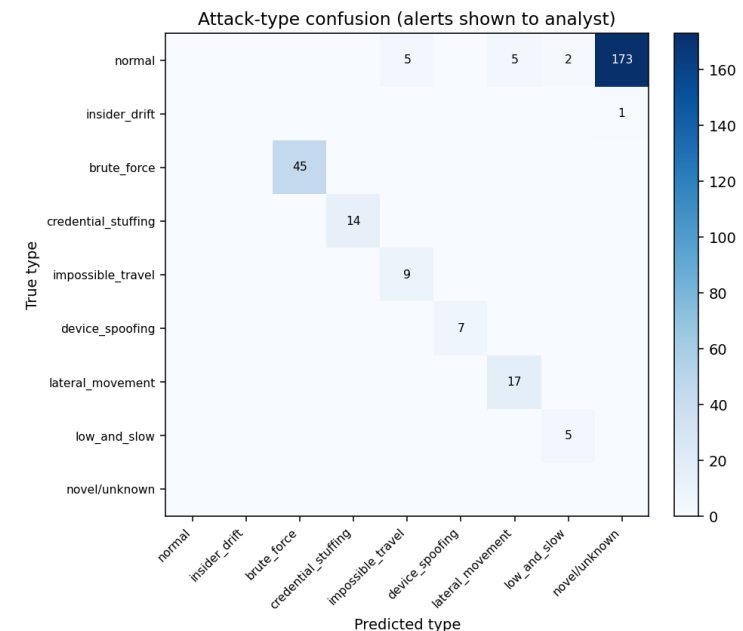
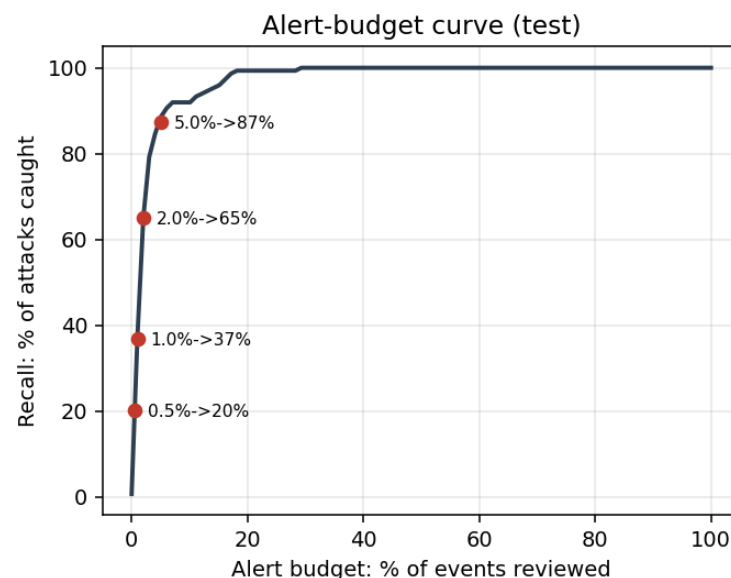
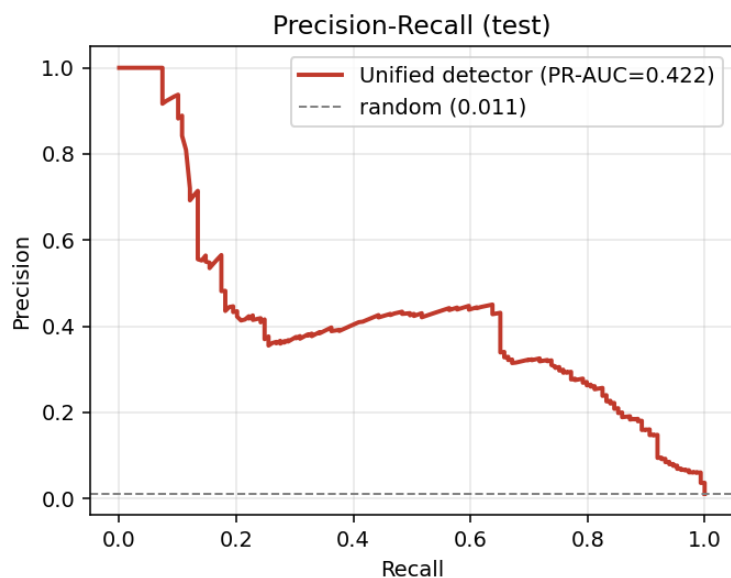
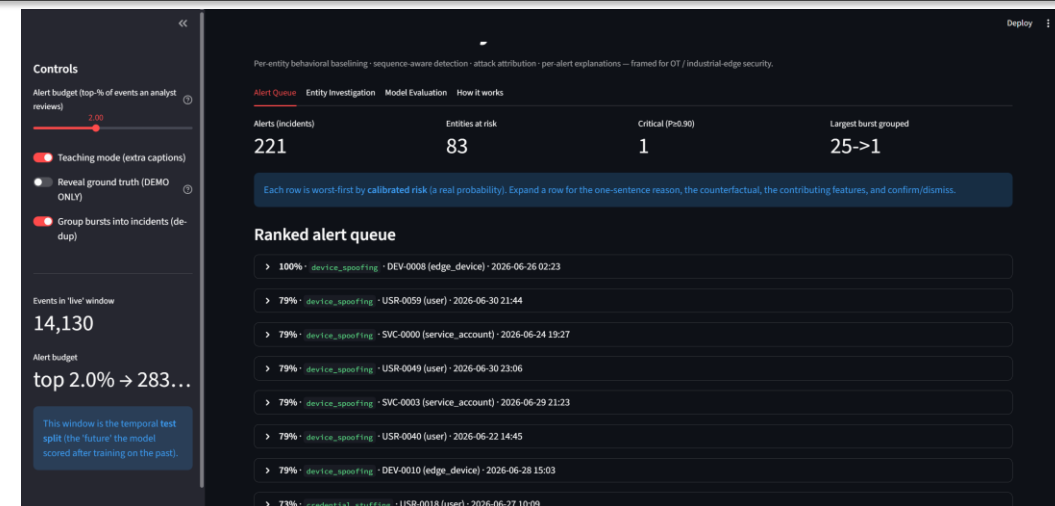
- Fewer missed intrusions
- Reduced alert fatigue
- Faster SOC triage
- Scalable for IT/OT environments

Challenges & Mitigation:

- **Low Precision** → Alert Budget + Risk Calibration
- **Concept Drift** → Adaptive Per-Entity Baselines
- **Novel Attacks** → Novel/Unknown Detection
- **Geo-Velocity Errors** → Counterfactual Explanations

ARTIFACTS

- **Source Code:** Modular Python implementation
 - Generator, Models, Dashboard, Evaluation & Reporting Modules
- **Project Deliverables:** Synthetic Security Dataset and Explainable AI Dashboard
 - Evaluation Report and Performance Visualizations.
- **Result:** figures below — PR-AUC · alert-budget · confusion-matrix.



RESEARCH AND REFERENCES

- Liu, Ting & Zhou — Isolation Forest, ICDM 2008.
- Sakurada & Yairi (2014); Malhotra et al. (2016) — autoencoder / LSTM-GRU reconstruction anomaly detection.
- Lundberg & Lee — SHAP unified feature attribution, NeurIPS 2017.
- Zadrozny & Elkan — isotonic calibration of classifier scores, 2002.
- MITRE ATT&CK — attack-technique taxonomy (brute force, lateral movement, exfiltration, valid accounts).
- UEBA (User & Entity Behavior Analytics) & SOC alert-fatigue practitioner literature; Honeywell Q4 brief.